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QUESTION 1 - WATER JUG PROBLEM
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Initial State
4-Gallon Jug = 0
3-Gallon Jug = 0

Step 1: Fill 4-Gallon Jug
4 0

Step 2: Pour into 3-Gallon Jug
1 3

Step 3: Empty 3-Gallon Jug
1 0

Step 4: Transfer remaining water
0 1

Step 5: Fill 4-Gallon Jug Again
4 1

Goal Achieved
2 Gallons remain in the 4-Gallon Jug

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QUESTION 2 - MARS ROVER
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PEAS Description
Performance : Safe Navigation
Environment : Mars Surface
Actuators   : Wheels, Robotic Arm, Drill
Sensors     : Camera, Temperature Sensor, Battery Sensor

Agent Model : Learning Agent

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Actions
- Move Forward
- Turn Left
- Turn Right
- Avoid Obstacle
- Collect Sample
- Recharge Battery
- Send Data

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QUESTION 3 - 8 QUEENS
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Q . . . . . . .
. . . . . Q .
. . . . Q . . .
. . . . . . . Q
. Q . . . . . .
. . . Q . . . .
. . . . . Q . .
. . Q . . . . .

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QUESTION 4 - OLA CAB BOOKING
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Customer Name : lakshmi
Source : chennai
Destination : madurai

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Available Cabs
1. Mini - Rs.150
2. Sedan - Rs.250
3. SUV - Rs.400

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Choose Cab : 2

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Booking Confirmed
Customer : lakshmi
Source : chennai
Destination : madurai
Cab : Sedan
Fare : Rs. 250

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Agent Model : Goal-Based Agent

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QUESTION 5 - UNIFORM COST SEARCH
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Optimal Path : S -> A -> C -> G
Total Cost : 4

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ASSESSMENT 1 COMPLETED SUCCESSFULLY
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